

JIHO KIM

C++ SYSTEMS · AUDIO / DSP · GPGPU · HAPTICS / XR

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PROFESSIONAL SUMMARY

Systems-focused C++ engineer specializing in cross-platform audio, real-time DSP and GPGPU acceleration. For over four years, I have designed and maintained C++20 audio middleware shipping on Windows, Linux and macOS with prebuilt Godot plugins—independently developing its GPGPU STFT kernels, audio/DSP runtime, IPC and CI/CD release pipeline. Published related research in APJCR and contributed a C++ bug fix to Meta's RocksDB. Currently researching haptic texture acquisition, modeling and vibrotactile rendering for Meta Quest 3 using OpenXR at Kyung Hee University's Haptics Lab.

AREAS OF EXPERTISE

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|---------------|-------------------------|-------------------------|--------------------------|
| • C++20 | • CMake | • Conan | • Clang |
| • MSVC | • CUDA | • OpenCL | • OpenMP |
| • OCCA / OKL | • GPU FFT / STFT | • Real-Time Audio / DSP | • miniaudio |
| • Faust | • OpenXR | • Meta Quest 3 | • Vibrotactile Haptics |
| • Cap'n Proto | • TCP / UDP / WebSocket | • shared-memory IPC | • RocksDB |
| • SQLite | • Godot 4 / GDEXTENSION | • CI/CD | • Cross-Platform Systems |
| • Linux | • Audio Middleware | • Signal Processing | |

PROFESSIONAL EXPERIENCE

M.S. Researcher — Haptics / XR, Haptics Lab, Kyung Hee University Mar 2025 to Present

Researching an end-to-end haptic texture acquisition, modeling and rendering workflow for Meta Quest 3 using OpenXR. Developing a low-cost force and vibration sensing probe, signal-processing and AR/LSF-based texture models, and real-time vibrotactile PCM rendering within a single mixed-reality session.

Project DJ Engine — Cross-Platform Audio Middleware Jun 2022 to Present

Solo open-source project across three repositories — core engine, Godot wrapper, prebuilt release — targeting Godot game developers. Built a cross-platform release pipeline that ships one C++20 codebase to Windows, Linux and macOS and auto-generates prebuilt Godot GDEXTENSION packages in CI/CD.

Undergraduate Researcher, CVLab, Hoseo Univ. Mar 2023 to Dec 2024

Worked on C++/computer vision research prototypes and experimental software development.

Developer — GPGPU / DSP, OCCA-STFT — Independent Open-Source Project Jul 2023 to Oct 2024

Developed a GPU-accelerated STFT implementation using OCCA/OKL, targeting portable execution across CUDA and OpenCL backends. Designed FFT/STFT compute kernels and integrated the library into a real-time audio/DSP workflow, with the work later supporting related research and Project DJ Engine.

EDUCATION

Kyung Hee University Mar 2025 to Present

M.S. in Computer Engineering

Hoseo Univ. Mar 2021 to Feb 2025

B.S. in Computer Engineering

LINKS

[GitHub](#)